Do Now · retrieval

Lesson 111 · OS Types, Boot Firmware, Drivers and VMs

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

In round robin, what happens when a running process's quantum expires but it is unfinished?

Q2

After pushing A then B onto a stack, which is popped first?

Q3

Why must a binary search operate on ordered data?

Q4

An OR gate has inputs 0 and 1. What is the output?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Match OS to use — Embedded/real-time emphasise constraints; multi-user and distributed coordinate shared work.

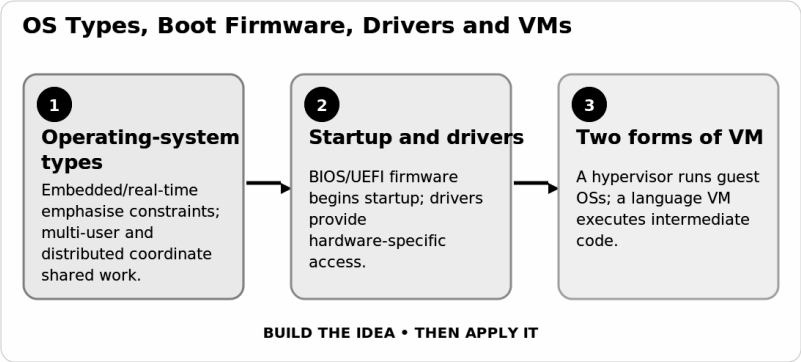
Core idea 2

Boot and control — BIOS/UEFI firmware begins startup; drivers provide hardware-specific access.

Core idea 3

Virtualise — A hypervisor runs guest OSs; a language VM executes intermediate code.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

real-time OS — System where meeting time constraints is essential.

BIOS/UEFI — Platform firmware that begins boot preparation.

hypervisor — Manages virtual machines running guest OSs.

device driver — Hardware-specific interface supplied to an OS.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

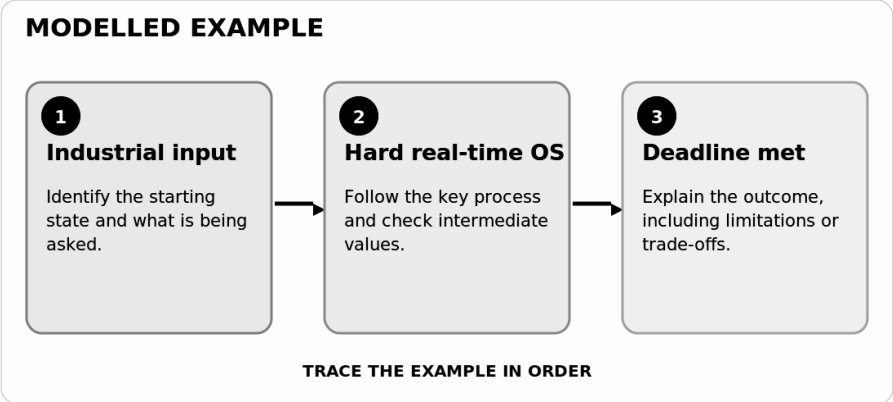
Problem

An industrial safety controller must respond within a guaranteed deadline. Choose an OS type and justify it.

Follow the worked method

1. Model answer — A hard real-time operating system is suited because missing a deadline is unacceptable; deterministic scheduling/latency matters more than high average throughput. An embedded implementation may also apply. — Check: link each claim to the scenario and use precise OCR terminology.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

What must we decide?
Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Explain two differences between multi-user and multitasking operating systems, giving examples. [6]

Question 2

Describe the roles of BIOS/UEFI firmware and a device driver during startup and use. [6]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Compare a hypervisor virtual machine with a bytecode virtual machine and explain a real-time deadline. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Every virtual machine must run a complete guest operating system.

Use key vocabulary from memory

1. Explain real-time OS in your own words.

2. Explain BIOS/UEFI in your own words.

3. Explain hypervisor in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.